

Julian Louis Ruocco

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Education

The Pennsylvania State University

B.S. Aerospace Engineering

Engineering Leadership Development Minor | Space Systems Engineering Certificate

May 2027

GPA 3.66/4.0

In Major GPA 3.82/4.0

Skills

Software & Simulation: MATLAB, Python, HTML, SolidWorks, Fusion, AutoCAD, STK, COMSOL, KiCad, LabView

Hardware & Fabrication: High Power Microwave Systems, Microwave-Assisted Pyrolysis, Arduino/Teensy Integration, Payload Assembly, Control Systems, RF Systems

Methodologies: Systems Engineering, MBSE, Digital Twin Modeling, Technical Proposal Writing, Mission Concepting

Work Experience

Mission Concept Intern, Space Dynamics Laboratory, Albuquerque NM

May - July 2026

- Developed a mission concept for a 3U CubeSat to detect, track, and identify ocean plastics in the Pacific Ocean.
- Simulated orbital parameters in STK to measure ground station passes, downlink capability, and orbit optimization.
- Built a Python script to compress satellite imagery into indexes capable of detecting ocean plastic.
- Utilized System's Engineering methodology to take an initial proposal from SCR to PDR.
- Worked with NASA and AFRL engineers to continually iterate the design to increase mission feasibility and relevance.

Student Space Programs Laboratory, State College PA

Laboratory Lead & Lead Systems Engineer

August 2025 - Present

- Developed a Digital Twin model linking COMSOL simulations, real time data from LabView, and CAD visualizations to our MBSE requirements for RECLAIM (NASA's LunaRecycle Challenge (Phase 2))
- Restoring and Upgrading Penn State's old UHF ground station while adding S band capability.
- Direct laboratory initiatives, grow the laboratory through strategic recruitment, and train 40+ student engineers.

Chief Engineer, NASA's LunaRecycle Challenge (Phase 2)

August 2025 - May 2026

- **Led 20 student engineers to win \$50,000 in the Milestone Round and secured a seat in the finalist round.**
- Created a proof-of-concept microwave-assisted plastic pyrolysis system proving feasibility.
- Utilized MBSE and regular reviews through SCR to CDR.

Research Lead, NASA's LunaRecycle Challenge (Phase 1)

December 2024 - May 2025

- Directed the research team for NASA's LunaRecycle Challenge (Phase 1), authoring key proposal sections.
- Created a MATLAB heat transfer model to gauge system temperature, improving testing efficiency by ~20%.

Member, Student Training Program

August 2023 - November 2024

- Applied systems engineering to design and build an atmospheric volatile gas detection payload using Arduino, Python, SolidWorks, and KiCad.

Research

Space Systems Researcher, Space Propulsion and Environments Laboratory, State College PA

May - August 2025

- Researched high power microwave systems for use on future long term lunar missions.
- Developed a technical proposal for a lunar recycling system which would fit within a centralized microwave network.
- Engineered systems design down to subsystem level and proposed novel catalyst selections for increasing methane generation to prove the feasibility of creating rocket propellant from plastic waste on future long term lunar missions.